

ENGINEERING PHYSICS LAB MANUAL

Course: Engineering Physics (22PH1005)

Summer term-Acceleration



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List of Experiments

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3	Determination of the energy band gap of a material	CO5
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6	Determination of Lattice Constant using P-XRD method	CO5

1. Determination of Solar Cell Characteristics

Aim: To draw the I-V Characteristics of a solar cell and to calculate the Fill factor Efficiency

Apparatus: Solar cell, Light source, Connecting wires, Voltmeter, Milliammeter, Potentiometer, Load resistance, etc.

Formula:

Fill factor efficiency = $P_m / I_{sc} \cdot V_{oc}$

Where P_m = Maximum Power Point

I_{sc} = Short Circuit Current

V_{oc} = Open Circuit Voltage

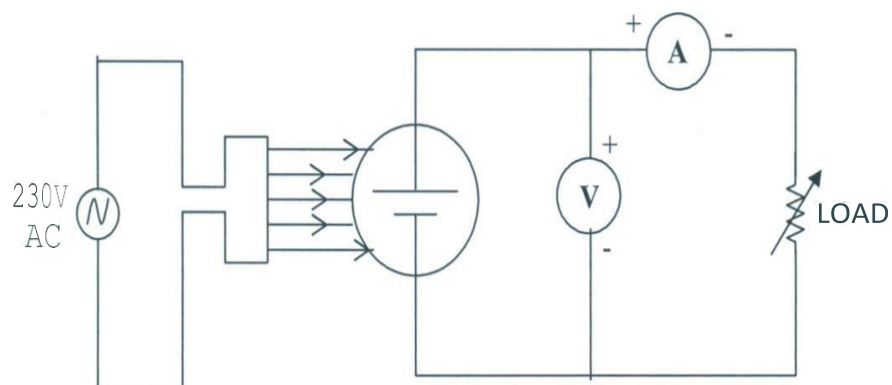
Procedure:

1. Complete the idealized equivalent circuit of solar cell connections.
2. Then place the light source S at 15 cm from the solar cell.
3. Adjust the potentiometer R_L until you obtain zero voltage on the voltmeter and maximum current in the ammeter (if it shows out of scale, then decrease the light intensity). This maximum current is called the short-circuit current I_{sc} .
4. Then, with the help of the potentiometer, increase the voltage in equal steps and note down the corresponding current till you get the maximum voltage on the voltmeter.
5. Now remove all the connections of the circuit and find out the open-circuit voltage (V_{oc}) [i.e., connecting the +ve of the cell to the +ve of the voltmeter and the -ve of the cell to the -ve of the voltmeter].
6. Repeat the experiment for other intensities, placing the light source at 20 cm and 25 cm, and plot the graphs between V and I.

Observation Table:

S.No.	Voltage (V) (Volts)	Current (I) (mA)	Power (VI)	Series Resistance $R = V/I$

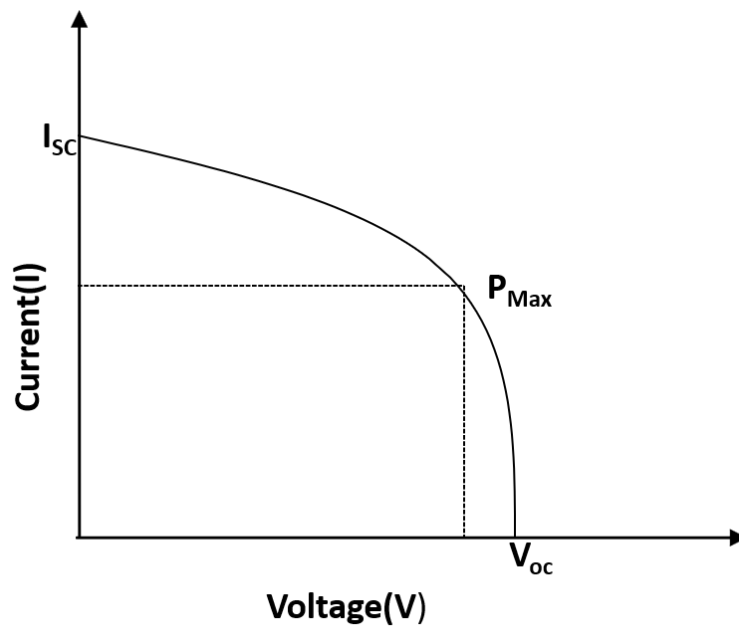
Circuit Diagram:



Precautions

1. The connections should be done correctly.
2. The light should fall on the solar cell properly.

Model graph:



Results:

- 1) V-I characteristics of a solar cell are studied.
- 2) Fill factor efficiency of the given solar cell is

Calculations:

2. PLANK'S CONSTANT BY PHOTOCELL

Aim: To determine Planck's constant from the stopping Potential measured at different Frequencies (wavelengths) of light.

APPARATUS:

The experiment set up consists of

1. Main unit
2. Vacuum photo tube
3. Light source
4. Color filters

Circuit diagram:

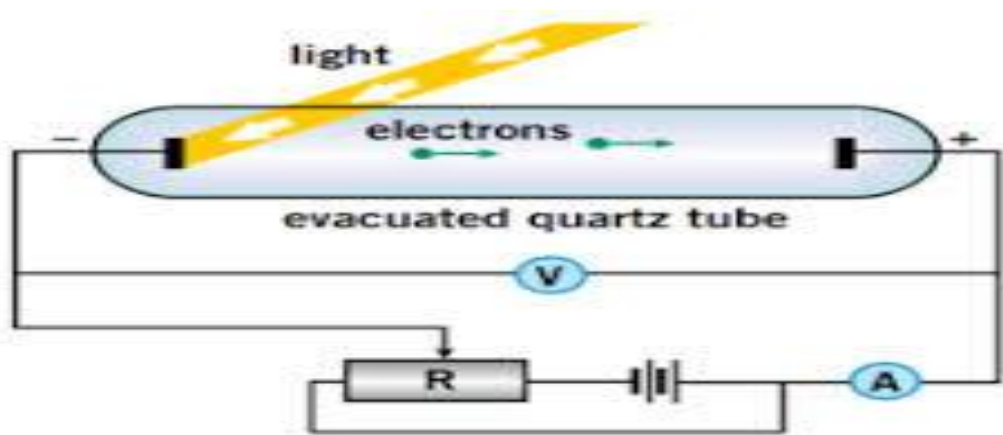


Fig. a: Photoelectric effect circuit connection

FORMULA:

$$h = e \times \frac{(V_2 - V_1)\lambda_1\lambda_2}{c(\lambda_2 - \lambda_1)} \text{Plank constant (Joule-Sec)}$$

h= Plank constant

V=Stopping Potential (calculated from the experiment for different filters)

C = Velocity of light (3×10^8 m/sec)

Λ = wavelength (given in table for different colors)

e = charge of electron (1.6×10^{-19} eV)

PROCEDURE

1. Make the connections as shown in fig 1. Switch light source to 100 Watt mode.
2. Keep the voltage control knob in its minimum position and switch on the main unit and light source at zero anode potential.
3. Arrange the light source and photo tube in such a way that light falls on the photo tube. The Distance between the photo tube and the light source is adjusted such that there is sufficient flow of current
4. Now place a color filter in the path of the light (A slot is given to place the filter).
- 5: After placing the filter, there will be some reading in the micro ammeter; this is called photo current at zero anode potential.
6. Now slowly increase the voltage till the current in the microammeter becomes zero; this voltage is called the stopping Potential V_0 . Wait for 2 minutes and note down this reading on the table.
7. Repeat the experiment with different color filters and note down their stopping Potential values.

OBSERVATIONS AND TABULATIONS:

S.No	Color	Stopping potential (V) in Volts	Wavelength(nm)	Plank constant (Joule-Sec) $h = e \times \frac{(V_2 - V_1)\lambda_1\lambda_2}{C(\lambda_2 - \lambda_1)}$
1	Blues	V1=	438 nm	

2	Yellow	V2=	578 nm	
3	Orange	V3=	598 nm	
4	Red	V4=	633 nm	
5	Green	V5=	565 nm	

The important observations on the photoelectric effect, which demand quantum theory for their explanation, are:

1. The Photoelectric effect is an instantaneous phenomenon. There is no time delay between the incidence of light and the emission of photoelectrons.
2. The number of photoelectrons emitted is proportional to the intensity of incident light. Also, the energy of emitted photoelectrons is independent of the intensity of incident light.
3. The energy of emitted photoelectrons is directly proportional to the frequency of incident light.

Result: The plank's constant value from experiment, $h =$ _____ J.S

Calculations:

3. Determination of the energy band gap of a material

Aim: To find the values of the band gap or energy gap of the semiconductor crystal.

Apparatus: p-n junction diode, DC regulated power supply, Voltmeter, micro-Ammeter, Beaker, Oil, Heater, and Thermometer.

Formula:

Energy gap of a p-n junction diode (E_g) = $2 K_B \times \text{Slope}$ *Joules*

$$k = 1.380649 \times 10^{-23} \text{ J K}^{-1}$$

$$E_g = (1.725 \times 10^{-4}) \times \text{Slope} \quad (eV)$$

Where,

$$\text{Slope} = \frac{\Delta \ln R}{\Delta \frac{1}{T}}$$

‘R’ is the resistance of a p-n junction diode at a particular temperature (*Ohms*).

‘T’ is the temperature of the oil bath (*K*)

Procedure:

The given p-n junction diode is connected in reverse bias and placed in an oil bath, as shown in Fig. 1. At room temperature, the saturation current is measured with a micrometer, and the corresponding bias voltage is also measured with a voltmeter. By maintaining the bias voltage at a constant value, the oil bath temperature is raised slowly, and the saturation current (*I*) values are recorded as a function of temperature in 5 °C increments. The oil bath heating is switched off when the temperature reaches 90°C, and the saturation current is recorded as the bath cools to near room temperature. The readings are tabulated.

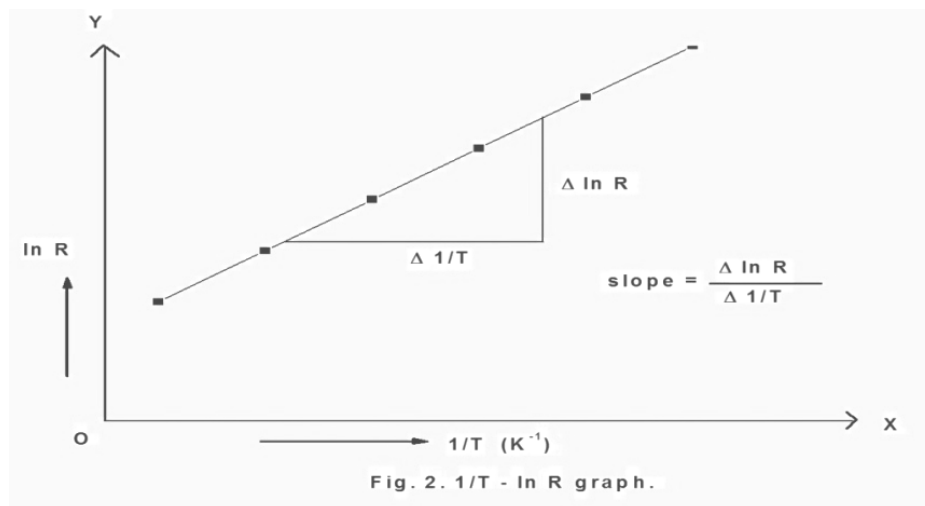
S. NO	TEMP (°C)	CURRENT ON COOLING	$R=V/I \times 10^6$	$\ln R$	$T=(273+t)$	$1/T \text{ K}^{-1}$
1	90					
2	85					
3	80					
4	75					
5	70					
6	65					
7	60					
8	55					
9	50					
10	45					

Precautions:

- 1) The diode has to be completely immersed in the oil.
- 2) The diode as well as the thermometer should not touch the walls of the beaker or heater coil.
- 3) In oil bath, thermometer and diode are approximately at same level.
- 4) Do not operate at temperatures beyond 90°C.
- 5) Avoid the high currents and loose connections.
- 6) In circuits, voltmeters must be connected in parallel to diode and ammeter must be connected in series combination.
- 7) Measurements should be taken without parallax error (*in case of analog meters*).

Graph: A graph is drawn between $\frac{1}{T} \times 10^{-3}$ and $\ln R$. The nature of the graph is a straight line as shown in fig. 2. Substitute the value of slope in the above formula and calculate the energy gap of the semiconductor material.

Model graph:



The graph is drawn between $\frac{1}{T} \times 10^{-3}$ and $\ln R$.

Result: The energy gap of the given p-n junction diode is _____
eV.

Calculation:

4. Determination of the dielectric constant of a given dielectric material

Aim: To study the temperature dependence of the dielectric constant of a ferroelectric substance

Apparatus: Dielectric constant apparatus, variable gang capacitor, test capacitor for solid dielectric material, connecting leads, and instruction manual.

Formula: Dielectric Constant, K, is given by

$$K = C1 - C2 / C1 - C3$$

C1 = capacity of standard variable capacitor at resonance (Maximum deflection).

C2 = capacity of standard variable capacitor at resonance (Maximum deflection), including the test capacitor with dielectric in it.

C3 = capacity of standard variable capacitor at resonance (Maximum deflection), including the test capacitor without dielectric in it.

Tabular Form:

S. N o	Thickness of dielectric material in mm	Value of variable capacitor at resonance when		
		Alone C1	With a test capacitor with dielectric in it) C2	With test capacitor(without dielectric in it) C3
1				
2				
3				
4				

Procedure:

1. Place the ferroelectric, which is in the form of a circular disc, between the plates of the parallel-plate capacitor. Reduce the spacing between the plates by gently rotating the screw until the ferroelectric sits snugly between the plates.
2. Connect the capacitor meter and measure the capacitance at room temperature.

Result:

dielectric constant of a ferroelectric = ----- K

Calculations:

5. Determination of particle size of lycopodium powder using laser.

Aim

To determine the particle size of microparticles (lycopodium powder) using a laser diffraction grating.

Apparatus Required

Laser source, fine microparticles having nearly the same size (say Lycopodium powder), Glass plate, Screen, and a meter scale.

Formula

Size of the microparticle using laser diffraction grating, $2d = \frac{n\lambda D}{x_n} m$

where n is the order of diffraction

λ is the wave length of laser light used in m (690 nm)

D is the distance between the glass plate and the screen in m

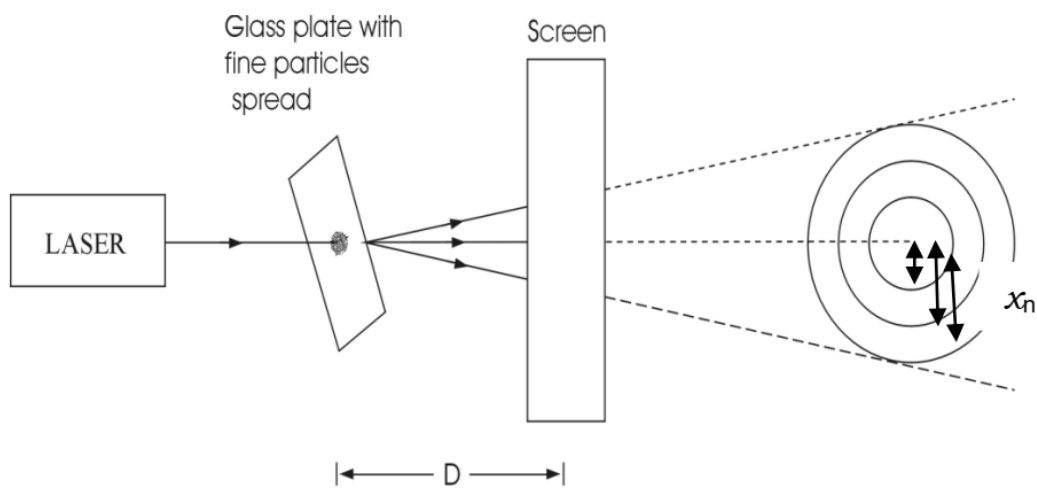
x_n is the distance between the central bright spot and the n^{th} fringe in m.

Procedure

When LASER is passed through a glass plate spread with fine micro particles, the beam gets diffracted by the particles and circular rings are obtained on the screen as shown in Fig. 1. By measuring the radii of the rings and the distance between the glass plate and the screen, the size of the particle can be determined.

1. Sprinkle the fine micro particles (Lycopodium) on the glass plate.
2. Mount the LASER source on a stand
3. Place a screen in front of the LASER source at some distance
4. Mount the glass plate on a separate stand and place it between the LASER source and the screen.

5. Switch ON the laser source and allow the beam to pass through the glass plate.
6. Adjust the distance (D) between the glass plate and the screen to get a clear circular fringe pattern (diffraction pattern) on the screen. The intensity is found to decrease from zeroth order (central spot) to higher orders.
7. Measure the distance (D) between the glass plate and the screen using metre scale.
8. Measure the distance (x_n) of the first order, second order and so on from the central bright spot (radii of the rings).
9. Repeat the experiment by varying the distance (D) between the glass plate and the screen and the readings are tabulated.



To determine the particle size

S. No.	Distance between the screen and the glass plate (D) $\times 10^{-2}$ m	Order of diffraction n	Distance between the central bright and n^{th} fringe (Δ) $\times 10^{-2}$ m	Particle size 2d $\times 10^{-5}$ m
1	1 2 3			
2	1 2 3			
3	1 2 3			

Mean 2d = -----

Result:

The size of the microparticle using laser diffraction grating, 2d =

Calculations:

6. Calculation of Lattice Cell Parameter - X-Ray Diffraction Method

AIM:

To calculate the lattice cell parameters from the powder X-ray diffraction data.

APPARATUS REQUIRED:

Powder X-ray diffraction diagram

PRINCIPLE: Crystalline substances act as X-ray diffraction gratings, producing constructive interference when Bragg's Law is satisfied, relating

FORMULA

For a cubic crystal,

$$\text{The lattice constant} \quad a = \frac{\lambda}{2 \sin \theta} \sqrt{h^2 + k^2 + l^2} \text{ \AA}$$

$$\text{The interplanar distance} \quad d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \text{ \AA}$$

Symbol	Explanation	Unit
a	Lattice constant	\AA
d	Interplanar distance	\AA
λ	Wavelength of the Cu K α radiation (1.5405 \AA)	\AA
(h k l)	Miller indices	-

PROCEDURE

From the 2θ values on a powder photograph, the θ values are obtained. The $\sin^2 \theta$ values are tabulated. From that the values of $1 \times \frac{\sin^2 \theta}{[\sin^2 \theta]_{\min}}$, $2 \times \frac{\sin^2 \theta}{[\sin^2 \theta]_{\min}}$, $3 \times \frac{\sin^2 \theta}{[\sin^2 \theta]_{\min}}$ are determined and are tabulated. The values of $3 \times \frac{\sin^2 \theta}{[\sin^2 \theta]_{\min}}$ are rounded to nearest integer, this gives the value of $(h^2 + k^2 + l^2)$. From these the values of (hkl) are determined from the Table.5.1. From the (hkl) values, the lattice parameters are calculated using the relation .

the wavelength of electromagnetic radiation to the diffraction angle and lattice spacing.

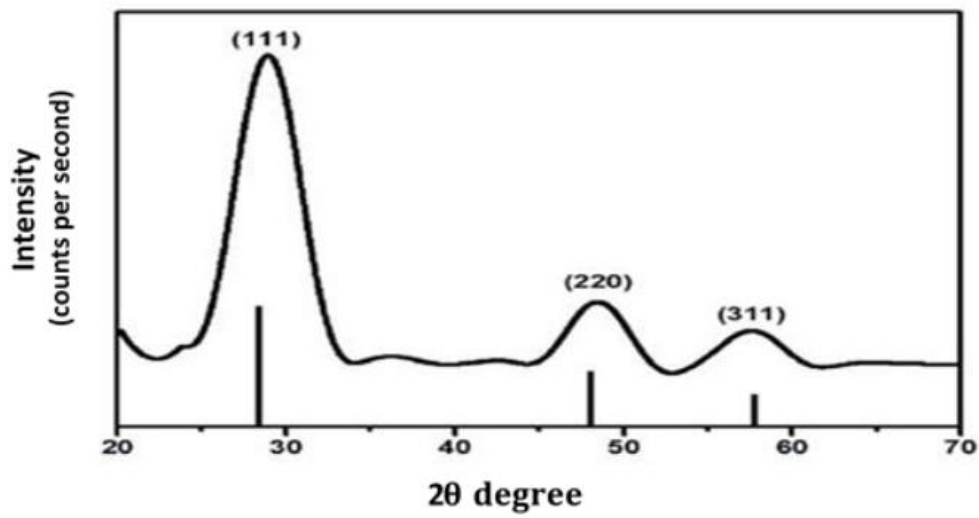


Fig 5.1 XRD pattern

h, k, l	$h^2 + k^2 + l^2$	h, k, l	$h^2 + k^2 + l^2$
100	1	300	9
110	2	310	10
111	3	311	11
200	4	322	12
210	5	320	13
211	6	321	14
220	8	400	16
221	9	410	17

Table 5.1 Value of $h^2 + k^2 + l^2$ for different planes

CALCULATION

The lattice parameter

$$a = \frac{\lambda}{2 \sin \theta} \sqrt{h^2 + k^2 + l^2} \text{ \AA}$$

The interplanar distance

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \text{ \AA}$$

Result: The lattice parameters are calculated theoretically from the powder X-ray diffraction pattern.